

Jiaming Deng

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Work Experience

TikTok

Nov.2023 - Present

Software Engineer

Dublin, Ireland

Working on video platform focus on storage middleware and Oncall AI Agent services. Used **Go, Python, MySQL, MongoDB, Redis, Object Storage, RocketMQ, Eino, LangGraph**

- **Object Cache Service:** Developed and maintained small object cache storage services, supporting 10M+ qps video frame downloads with 82%+ hit rate.
- **Edge Storage Service:** Developed and maintained video redundancy temporary storage services, supporting 12k+ qps video upload in edge clusters.
- **Cache Addressing Service:** Designed and implemented a high-performance, decentralized cache addressing architecture using consistency hashing with virtual nodes and Gossip-based topology discovery, replacing a single central lookup service so that cache clients route requests directly to target nodes, significantly reducing metadata lookup latency and improving read/write throughput
- **Oncall AI Agent:** Built comprehensive intelligent operations AI Agent from scratch with multi-agent architecture centered on supervisor that selects specialized sub-agents equipped with different MCP tools and prompts, enhancing execution accuracy and preventing tool misuse. Applied to 5 products in TikTok video platform for automated RCA and self-recovery.
- **MCP / Skill Development:** Researched and developed extensive MCPs and Skills by integrating TikTok's private cloud services and internal systems including k8s, configuration center, metrics, grafana, object storage, redis, and many customized video platform-specific MCP tools. Grafana MCP tools has been published to TikTok's internal MCP market with widespread adoption.

Trinity College Dublin

Jun.2023 – Oct.2023

Software Engineer - Research Assistant

Dublin, Ireland

Multiple projects involving microservices and fast-paced research-facing requirements. Used **JavaScript, React, MongoDB**

- Maintained and optimized the algorithm for fetching data required by researchers by reducing redundant data and multiple stream pipelines and Gzip, resulting in a 30% improvement in obtaining data.
- Developed a well-tested web application for researchers to view, analyze and export neural scientific data and for developers to view well-documented webserver endpoints.
- Built a permission-managing web server and web application for researchers to control their access to citizens' experimental data.

GrabVoice

Mar.2023 – Jun.2023

Software Engineer Intern

Dublin, Ireland

Worked at a fast-growing **Start-up**, responsible for building its main service including distributed systems and microservices. Used **Go, Gin, MySQL, MongoDB**.

- Implemented an authentication and authorization system to control users' different permission to the service.
- Developed strategies to remove Grabvoice's expired deals through either logical deletion or permanent deletion.
- Deployed and managed projects on AWS EC2 instances and Implemented AWS S3 as the primary database storage solution for the project, ensuring efficient performance and scalability.

Education

Trinity College Dublin

Sep.2022 – Jun.2023

Master of Science in Computer Science

Dublin, Ireland

Xi'an Jiaotong University

Sep.2018 – Jul.2022

Bachelor of Engineering in Computer Science

Xi'an, China

Technical Skills

Languages: Go, Java, Python, JavaScript, C++

LLM Services&Tools: Eino, LangChain, LangGraph, Codex, Claude Code

Backend Frames: Gin, Spring Boot, Express, KiteX, Hertz

Database&Storage: Redis, MySQL, MongoDB, RocketMQ, Object Storage